

Floatsam

Overall progress, the decisions behind it, and the open risks — a single source of truth across product, engineering, and planning.

Date 2026-07-11 Live nikhilprakash24.github.io/floatsam Health **Green**
Stack **Phaser** · **TypeScript** · **Vite**

EXECUTIVE

§1 orientation, §2 progress, §6 roadmap.

ENGINEERING

§3 pillars, §4 decisions, §5 risks.

PLANNING

§2 status, §4 decisions, §6 what's next.

§1 Orientation

Floatsam is a shipped, live underwater arcade game built on an unusual foundation: the entire game is a deterministic, headless simulation with fairness proven by the thousands of runs — not by playtest vibes.

You tap or dive a sea creature through gaps in a reef. Under the hood, a pure-TypeScript fluid-physics core (zero rendering dependencies) runs a fixed-timestep simulation seeded so every run is exactly reproducible. That property is the project's spine: it lets us freeze a "golden" reference run, machine-verify fairness across 80,000 seeded runs, and add features behind hard regression gates.

§2 Progress by track

TRACK	STATUS	DETAIL
Core game	SHIPPED · LIVE	3 modes (Classic / Dive / Power Dive), 4 creatures with trait cards, tempo wrapper, per-mode/creature best scores.
Platform	SHIPPED	PWA (installable, offline), GitHub Actions CI, Pages deploy on every push. Accessibility & best-practices 100.
Finale graphics	SHIPPED · LIVE	Animated creatures, wave title, Fable Currents lab, distant seamount backdrop (replaced light shafts), rare whale fly-by.
Currents program	BUILT · PR OPEN	C1–C6 on a branch: 20-pattern catalog, scrolling current cells in Power Dive, rare whole-screen surge, live-tuning sandbox. Default-off.
Currents → public	PENDING	Full fairness sweep, a player-facing entry, and feel-tuning before promotion.

§3 Engineering pillars

Three properties make the codebase unusually safe to extend, and are worth protecting in every future decision:

- **Determinism.** A seeded, fixed-timestep, Phaser-free simulation core. Same seed ⇒ same run, every platform. Enables replay, golden masters, and headless fairness testing.
- **Fairness by construction.** Difficulty is bounded by derived reachability clamps and (for currents) a budget clamp anchored to the weakest creature — so "is this beatable?" is answered by math and a sweep, not opinion.
- **The golden master.** Classic (Seal) is tuning-locked to a frozen trajectory fixture; any change that would perturb it fails CI. New systems land *behind* this gate, default-off, proven bit-identical.

This is why the entire currents program — a large gameplay system — landed with **zero changes to the shipped game** and could ship as a reviewable branch with no blast radius.

§4 Key decisions

DECISION	RATIONALE
Classic is tuning-locked	A stable reference the whole test apparatus hangs on; changes need an ADR + fresh sweep.
A mode = data + input policy + field factory	New modes (Dive, Power Dive) add zero risk to existing ones; the pattern scales.
Power Dive introduced free-x movement	The deliberate seam through which lateral currents can matter — designed before currents existed.
Currents = scrolling cells + rare surge	A whole-screen field collapses to one column in a fixed-x scroller; localized cells are the right unit. (ADR-014, pending.)
Currents ship opt-in, default-off	Lets a large feature land without touching the golden master or fairness matrix.
Design forks as sandbox variables	Intensity, density, rarity, mode-coupling are knobs, so decisions become tuning, not debate.

§5 Risk register

#	RISK	SEV	MITIGATION / STATUS
R1	Currents fairness proven by 60-seed spot check, not the full 80k sweep	MED	Budget clamp bounds per-frame force; full sweep + currents golden is the tracked gate before promotion
R2	Currents difficulty escapable but not yet feel-tuned	MED	Sandbox built for exactly this; defaults deliberately gentle
R3	No player-facing currents entry yet	LOW · INTENDED	Deep-links + dev scenes today; menu/mode entry scoped next
R4	Single-maintainer bus factor; large tacit context	MED	Extensive in-repo docs, ADRs, plan-of-record, and these reports capture the "why"
R5	Perf headroom (Phaser boot dominates a throttled Lighthouse run)	LOW	a11y/best-practices 100; deferred-boot optimization logged, not blocking
R6	Fable model window closing (authoring velocity)	LOW	Work is committed, gated, and documented so it resumes cleanly

§6 Roadmap — what's next

- **Now:** merge the currents branch (zero-risk) to make the Catalog & Sandbox reachable at the live URL.
- **Next:** feel-tuning pass in the Sandbox; choose & build the player-facing currents entry (toggle vs. standalone mode).
- **Then:** full 80k currents fairness sweep + a frozen currents golden; ratify ADR-014.
- **Later:** graduate candidate catalog patterns as each clears the sweep; distribution polish (store packaging, perf boot).